Sperm Whale Coda Communication: Acoustic Structure, Turn-Taking, and Sequential Response Patterns

Mansur Gapar
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Abstract

Sperm whales communicate using patterned sequences of clicks known as codas, but understanding how these signals are organized during interaction remains challenging. This study combined acoustic analysis and sequential modeling to investigate whether specific coda responses occur non-randomly during cross-speaker exchanges. Publicly available sperm whale datasets derived from the Dominica Sperm Whale Project were used to analyze four 5-click coda types. Rhythm and total duration were evaluated using clustering and supervised classification, and dialogue-level data were analyzed for turn-taking and sequential response structure. The selected coda types were distinguished with a mean balanced accuracy of 0.997 under grouped cross-validation. Speaker-change gaps were substantially shorter than same-speaker continuations, with median across-file gaps of approximately 1.04 s and 3.64 s, respectively. The association between preceding and response coda types was stronger than expected under a repertoire-preserving permutation model (observed mutual information = 0.195; permutation $p \approx 0.0002$). The strongest enriched transitions were concentrated within the 5R coda family, including 5R1 → 5R2, 5R2→ 5R1, and 5R2 → 5R2. These results do not determine the semantic meaning of individual codas, but they identify reproducible interaction patterns that may help guide future behavioral studies of sperm whale communication.

Keywords: sperm whale; coda; bioacoustics; machine learning; turn-taking; sequential analysis

1. Introduction

Sperm whales (*Physeter macrocephalus*) are highly social animals that communicate using short sequences of clicks known as **codas**. A coda consists of a small number of clicks arranged with characteristic timing, and different timing patterns are traditionally classified as different coda types. These vocalizations are produced primarily during social interactions and are considered fundamental units of sperm whale communication.

Previous research has shown that coda communication contains structure at several levels. Sperm whales can coordinate the timing and type of codas during vocal exchanges, and matching of previously produced codas has been observed more often than expected by chance (Schulz et al., 2008). More recently, Sharma et al. (2024) showed that codas contain a richer acoustic organization based on rhythm, tempo, rubato, and ornamentation, demonstrating that sperm whale vocalizations are more combinatorial and context-sensitive than previously recognized. Automated analyses have also reported synchronization between interacting whales in both coda type and transmission timing (Gubnitsky et al., 2025).

These findings raise a further question: does the specific type of coda produced by a responding whale depend on the coda that immediately preceded it, beyond differences in the types of codas that individual whales tend to produce?

In this study, acoustic and sequential analyses were combined to investigate this question using publicly available datasets derived from recordings collected by the Dominica Sperm Whale Project. First, selected 5-click coda types were compared using their rhythmic structure and overall tempo, and their acoustic distinguishability was evaluated using clustering and supervised classification. The analysis then examined temporal turn-taking and adjacent cross-speaker coda transitions in dialogue recordings. Finally, repertoire-preserving permutation tests and robustness analyses were used to determine whether particular response patterns occurred more frequently than expected from local whale repertoires alone.

The aim was not to assign semantic meanings to individual codas, but to identify reproducible relationships between signals that may help narrow the search for communicative function. Such structured response patterns may highlight specific coda exchanges that future studies can investigate together with behavioral and social context.

2. Materials and Methods

2.1 Data Sources

The two datasets analyzed in this study were obtained from the official companion repository for Sharma et al. (2024), which contains the data and code associated with Contextual and Combinatorial Structure in Sperm Whale Vocalisations. The datasets are based on recordings collected by the Dominica Sperm Whale Project. The labeled coda dataset contained 8,719 annotated codas and included information such as coda type, total duration, number of clicks, inter-click intervals (ICIs), vocal clan, social unit, and individual identity. These data were used to examine acoustic structure and to train models for distinguishing selected coda types.

A second dialogue-level dataset contained temporally ordered coda events with speaker identities, timing information, click counts, and acoustic measurements. This dataset was used to investigate turn-taking and sequential response patterns between whales. Together, the two datasets allowed acoustic structure to be linked with the temporal organization of communication.

2.2 Data Preparation

The labeled coda dataset was first restricted to the EC-1 vocal clan to reduce variation associated with different cultural vocal repertoires. Entries labeled as noise were removed before further analysis. After cleaning, the dataset contained 7,268 codas representing 24 coda types.

The main acoustic analysis focused on four selected 5-click coda types: **1+1+3, 2+3, 5R1, and 5R2**. These categories were chosen because they contain the same number of clicks, allowing their internal timing patterns to be compared using the same four inter-click intervals. The resulting subset contained 5,448 codas.

For each 5-click coda, the four inter-click intervals were normalized by dividing each interval by the sum of the four intervals. This produced relative timing features that describe the internal rhythmic pattern independently of the overall coda duration.

2.3 Acoustic Feature Representation

A sperm whale click is a brief acoustic pulse, and a coda is a stereotyped sequence of clicks used during social communication. The timing between two consecutive clicks is referred to as the inter-click interval (ICI). For a 5-click coda, four ICIs describe the temporal spacing between the five clicks.

In this study, acoustic structure was represented using two complementary types of features: **rhythm** and **tempo**. Rhythm was described using the four normalized ICIs. Each ICI was divided by the sum of all four intervals in the coda, producing relative timing values whose sum equals one. This normalization reduces the influence of overall production speed and emphasizes the internal temporal pattern of the coda.

Tempo was represented by the total duration of the coda. While normalized ICIs capture how the timing is distributed within a coda, total duration captures how quickly or slowly the complete click sequence is produced. Using both feature types therefore allowed rhythmic shape and overall speed to be analyzed separately and in combination.

2.4 Unsupervised Clustering

K-means clustering was used to test whether the selected coda types formed natural acoustic groups without providing the algorithm with the known coda labels. K-means is an unsupervised learning algorithm that assigns observations to a predefined number of clusters, denoted by k, based on similarity in their feature values.

Because the four selected coda types were strongly imbalanced in the original dataset, balanced samples were created by randomly selecting 77 examples from each type, corresponding to the size of the smallest class. This produced 308 codas per balanced sample. The procedure was repeated across 50 random seeds to reduce dependence on a single random sample.

Three feature representations were compared: normalized rhythm features alone, total duration alone, and rhythm combined with duration. Features were standardized before clustering so that variables measured on different scales contributed comparably to the distance calculations. K-means models were evaluated for values of k from 2 to 6.

Clustering quality was assessed using the silhouette score (Rousseeuw, 1987), which measures how well each observation fits within its assigned cluster relative to neighboring clusters. Agreement between the unsupervised clusters and the known coda-type labels was additionally evaluated using the Adjusted Rand Index (ARI; Hubert & Arabie, 1985) and Normalized Mutual Information (NMI). ARI and NMI were used only for evaluation and were not provided to the clustering algorithm during clustering.

2.5 Coda-Type Classification and Grouped Validation

To test whether the four selected coda types could be reliably distinguished from their acoustic features, a supervised classification model was trained using normalized ICIs together with total coda duration. A logistic regression classifier was used within a pipeline that standardized the input features before

classification (Pedregosa et al., 2011). Logistic regression is a supervised learning method that estimates the probability that an observation belongs to each predefined class.

Because the four coda types were strongly imbalanced, class weights were balanced during model fitting so that rare classes contributed more equally to the classification objective.

Model performance was evaluated using five-fold Stratified Group K-Fold cross-validation. Instead of randomly separating individual codas, observations were grouped using the combination of Date and Unit, which served as a proxy for recording context. Codas belonging to the same group were kept entirely within either the training or test portion of a fold, preventing overlap between related recording contexts.

Performance was evaluated using overall accuracy, balanced accuracy, precision, recall, and F1-score. Balanced accuracy was included because it gives equal importance to each class and is therefore less sensitive to differences in class frequency.

2.6 Dialogue Timing Analysis

To examine the temporal organization of sperm whale exchanges, coda events in the dialogue dataset were first ordered by audio file and event time. For each event, the identity of the previous whale, the previous event time, and the duration of the previous coda were recorded.

A response gap was calculated as the time between the end of the previous coda and the start of the following coda. Consecutive events were then classified as either **speaker-change transitions**, in which the following coda was produced by a different whale, or **same-speaker transitions**, in which the same whale produced consecutive codas.

Response-gap distributions were compared between these two transition types. Because recording conditions and interaction patterns varied across audio files, median response gaps were also calculated separately within each audio file. Only files containing both transition types were included in the paired comparison.

The paired within-file medians were compared using a one-sided Wilcoxon signed-rank test. This non-parametric test was used to determine whether same-speaker continuation gaps were systematically longer than speaker-change gaps without assuming normally distributed timing differences.

2.7 Cross-Speaker Transition Analysis

To examine whether the type of coda produced by one whale was associated with the type produced next by another whale, the coda-type classifier was applied to 5-click events in the dialogue dataset. The same four normalized inter-click intervals and total duration used during classifier training were calculated for each dialogue coda.

Classification confidence was estimated from the maximum predicted class probability. To reduce uncertain assignments, a coda was retained as one of the four known types only when the maximum

predicted probability was at least 0.95 and all five acoustic features fell within the minimum and maximum ranges observed in the labeled training dataset. Codas that failed either criterion were labeled **Unknown**. This range check was used as a simple safeguard against assigning known labels to acoustically unusual observations.

The predicted labels were then mapped back onto the complete chronological dialogue sequence. Non-5-click codas were retained in the timeline even though they were not assigned one of the four analyzed coda types. This ensured that two 5-click codas were considered consecutive only when no intervening coda event occurred between them.

A signal-response pair was defined when two adjacent events were both assigned one of the four known coda types, were produced by different whales, and had a response gap between −2 and 5 seconds. Negative response gaps were allowed because some consecutive codas partially overlapped in time. These filtered pairs formed the dataset used for the cross-speaker transition analysis.

2.8 Permutation Testing and FDR Correction

To test whether response coda types were associated with preceding coda types more strongly than expected by chance, permutation tests were performed on the cross-speaker signal-response pairs.

First, the overall dependence between previous and response coda types was quantified using mutual information. Mutual information measures how much knowing one categorical variable reduces uncertainty about another. The observed value was compared with a null distribution generated by randomly shuffling response labels within each AudioFile × responding Whale group. This procedure preserved each whale's local response repertoire within a recording while breaking the sequential relationship between the preceding and following coda types.

A total of 5,000 permutations were used. The permutation p-value was calculated as the proportion of null values greater than or equal to the observed statistic, using a +1 correction in both the numerator and denominator.

Transition-specific permutation tests were then performed for all 16 possible previous-type → response-type combinations. For each transition, the observed count was compared with counts obtained under the same within-group shuffling procedure. These tests were one-sided and evaluated enrichment, meaning whether a transition occurred more often than expected under the null model.

Because multiple transition-specific hypotheses were tested, p-values were adjusted using the Benjamini-Hochberg false discovery rate (FDR) procedure (Benjamini & Hochberg, 1995). This correction reduces the expected proportion of false-positive findings among transitions identified as significant.

2.9 Robustness Analysis

Several robustness checks were performed to determine whether the main sequential findings depended strongly on particular analytical choices or a small number of recordings.

First, the cross-speaker transition analysis was repeated using maximum response-gap windows of 3, 5, and 8 seconds. This tested whether the main transition patterns remained stable when the temporal definition of a response was made more or less restrictive.

Second, the distribution of key 5R transitions across audio files was examined. For each transition, the total number of observations, the number of audio files containing that transition, and the largest contribution from any single audio file were calculated. This was used to assess whether an apparent effect was dominated by one recording.

Third, the same concentration analysis was performed at the whale-pair level. An unordered whale-pair identifier was created within each audio file, and transition counts were summarized across these pair contexts. This tested whether the main patterns were concentrated within a single interacting pair or appeared across multiple pair contexts.

These robustness analyses were used to evaluate the stability and generality of the observed transition structure rather than to establish additional semantic interpretations of the codas.

3. Results

3.1 Acoustic Structure

The four selected 5-click coda types differed in both total duration and internal rhythmic structure. Mean coda duration was approximately 1.073 s for 1+1+3, 0.655 s for 2+3, 0.330 s for 5R1, and 0.510 s for 5R2.

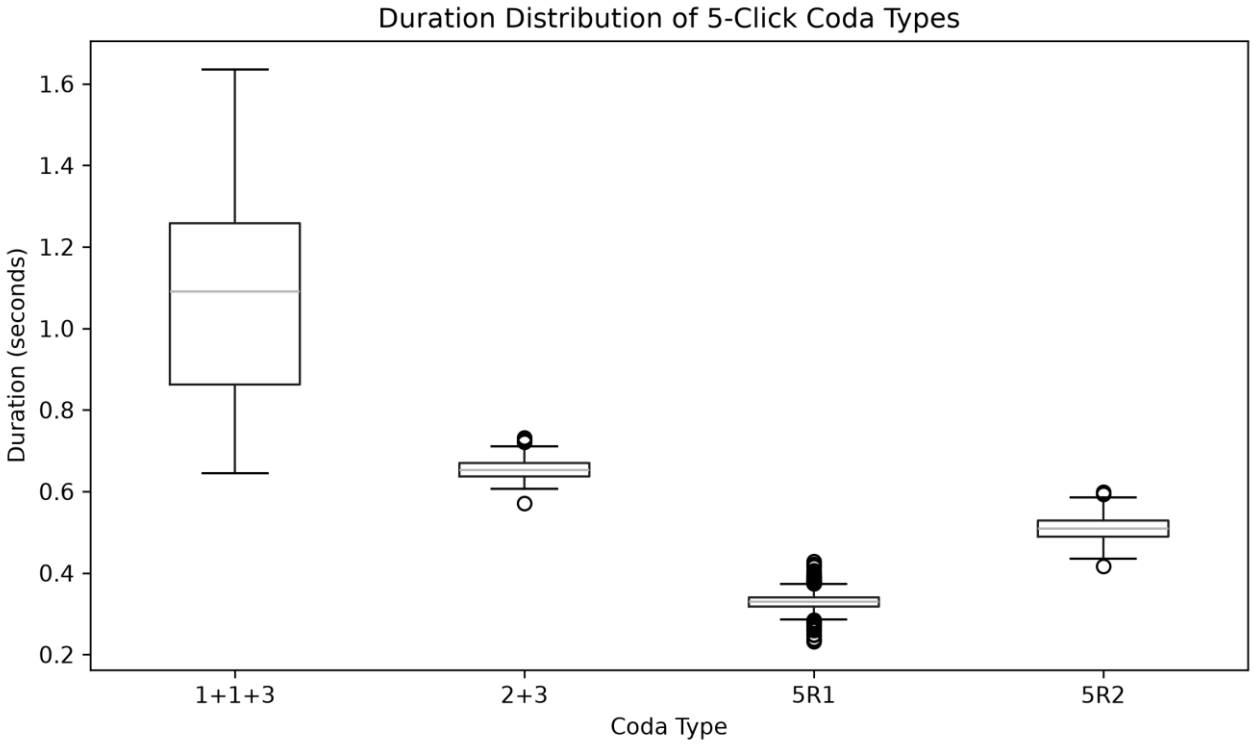


Figure 1. Duration distribution of the four selected 5-click coda types. Boxplots show differences in total coda duration across 1+1+3, 2+3, 5R1, and 5R2.

The normalized inter-click interval profiles also differed across coda types, indicating distinct internal timing patterns. The 1+1+3 and 2+3 codas showed particularly different rhythmic profiles, while 5R1 and 5R2 were more similar in relative rhythm but differed substantially in overall duration.

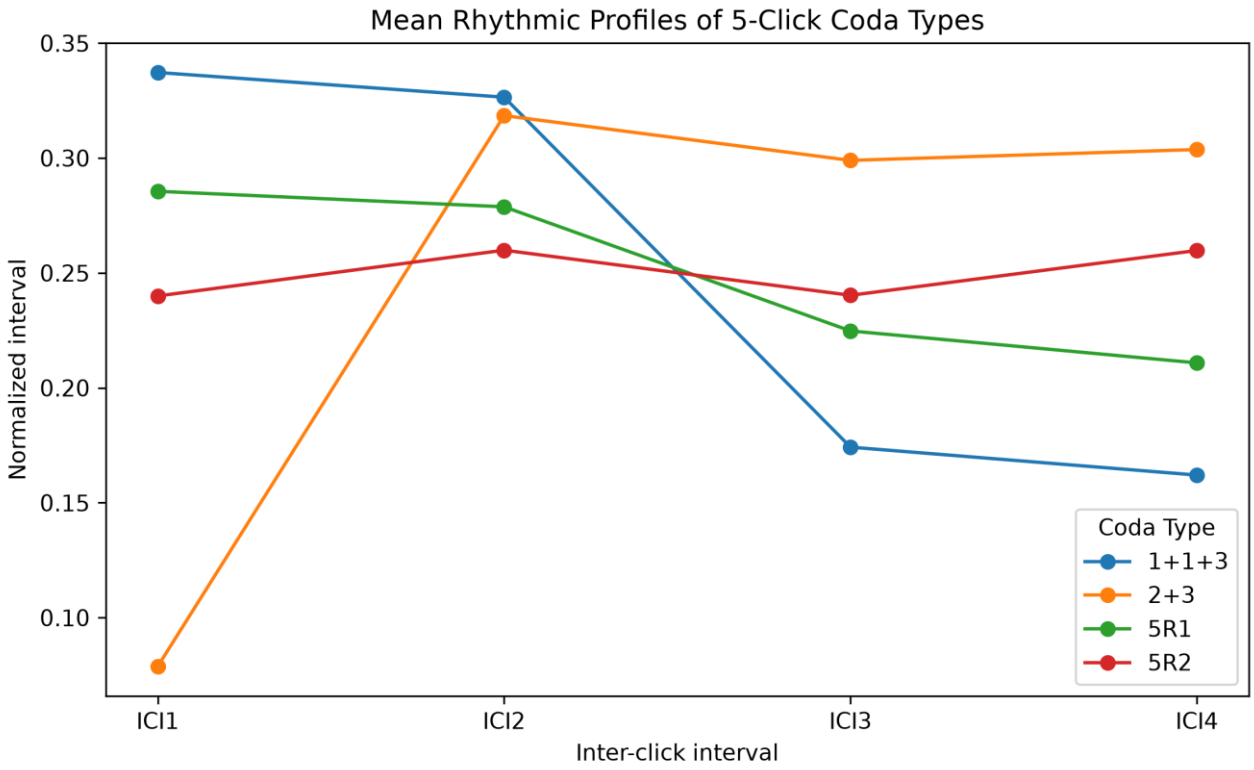


Figure 2. Mean rhythmic profiles of the four selected 5-click coda types**.** The four normalized inter-click intervals (ICI1–ICI4) summarize the relative timing pattern within each coda type.

Together, these results indicate that both relative rhythm and overall tempo contribute to the acoustic distinction between the selected coda categories.

3.2 Unsupervised Clustering

Repeated K-means clustering showed that the recovered structure depended on which acoustic features were used. When four clusters were evaluated, rhythm alone produced a mean silhouette score of 0.502, an Adjusted Rand Index (ARI) of 0.601, and a Normalized Mutual Information (NMI) score of 0.718.

Tempo alone produced stronger separation at four clusters, with a mean silhouette score of 0.673, ARI of 0.717, and NMI of 0.804. The combined rhythm-and-tempo representation produced the highest agreement with the known four coda categories, with a mean ARI of 0.801 and mean NMI of 0.823, although its silhouette score was lower at 0.469.

Across the tested values of k, the results showed that the most geometrically compact clustering solution was not always the solution that best matched the known coda-type labels. This suggests that coda-type structure is influenced by multiple acoustic dimensions rather than forming four perfectly separated geometric clusters.

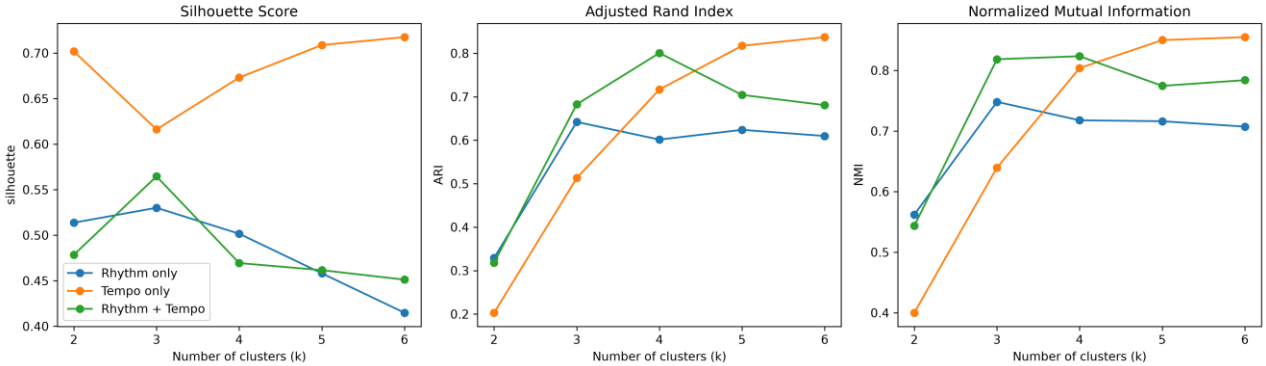


Figure 3. Clustering performance across feature representations and numbers of clusters. Silhouette score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) are shown for rhythm-only, tempo-only, and combined rhythm-plus-tempo representations.

3.3 Classification Performance

The supervised classifier distinguished the four selected coda types with very high accuracy. Across five folds of stratified grouped cross-validation, mean overall accuracy was 0.996 and mean balanced accuracy was 0.997. The standard deviation of balanced accuracy across folds was 0.0024, indicating highly consistent performance.

No Date + Unit group appeared in both the training and test sets within any fold, resulting in a maximum group overlap of zero. This reduced the risk that model performance was driven by repeated observations from the same recording context.

Class-wise performance was also high. The F1-scores were 0.999 for 1+1+3, 0.994 for 2+3, 0.995 for 5R1, and 0.965 for 5R2. These results indicate that the selected coda categories were acoustically stable and could be reliably distinguished using normalized rhythm and total duration.

This classification result was used to support the transfer of coda-type labels to 5-click events in the dialogue dataset.

3.4 Turn-Taking Timing

Speaker-change transitions occurred after substantially shorter gaps than same-speaker continuations. Across all events, the median speaker-change gap was approximately 0.73 s, compared with approximately 3.52 s for same-speaker transitions.

To reduce the influence of differences between recordings, median response gaps were also calculated separately within each audio file. Among the 40 audio files containing both transition types, the median

speaker-change gap across files was approximately 1.04 s, whereas the median same-speaker gap was approximately 3.64 s.

The within-file difference was positive in every analyzed audio file. A one-sided Wilcoxon signed-rank test confirmed that same-speaker gaps were significantly longer than speaker-change gaps ($W = 820$, $p \approx 9.09 \times 10^{-}13$).

These results indicate a consistent temporal turn-taking pattern in which a different whale typically responds sooner than the same whale produces its next coda.

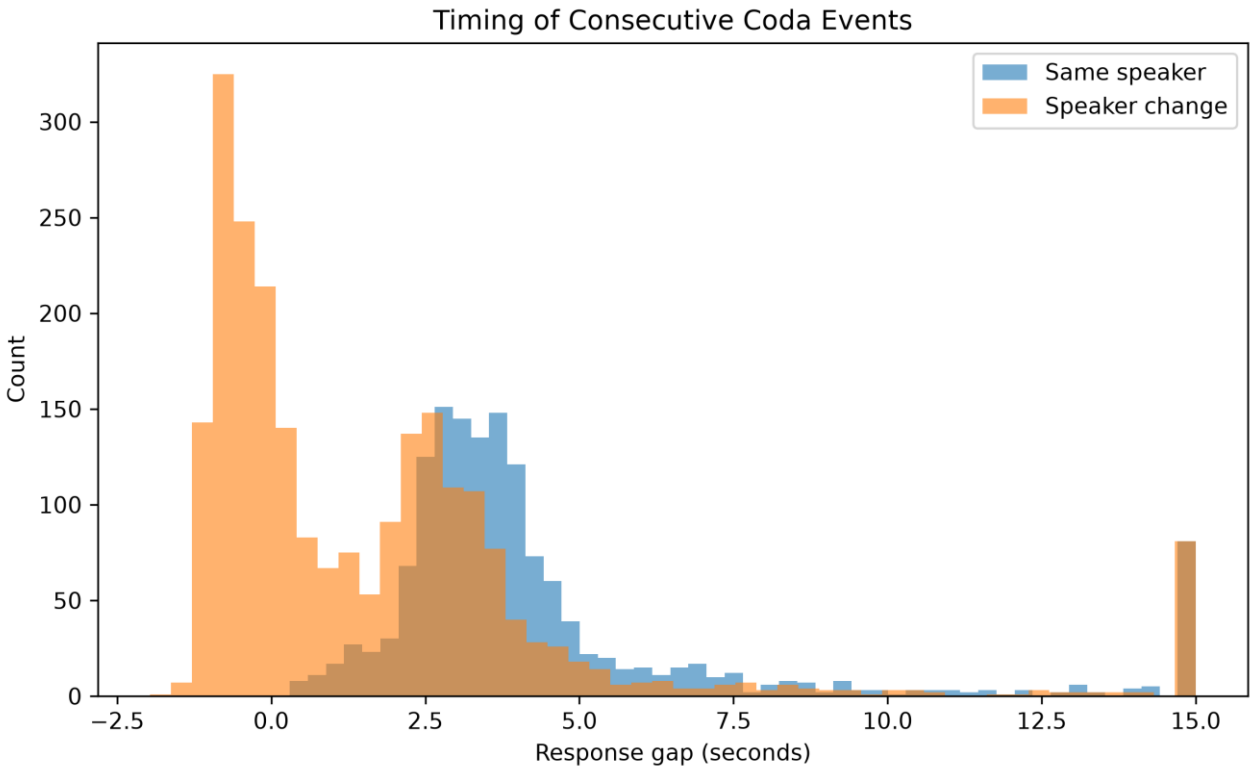


Figure 4. Distribution of response gaps for consecutive coda events**.** Speaker-change transitions were concentrated at shorter response gaps than same-speaker continuations.

3.5 Sequential Response Structure

After applying the confidence and acoustic-range filters, the dialogue analysis produced 1,224 high-confidence adjacent cross-speaker signal-response pairs within the 5-second response window.

The transition matrix showed that response frequencies depended strongly on the preceding coda type. For example, 5R1 was followed by another 5R1 in 119 cases, while 5R2 was followed by 5R1 in 13 cases and by another 5R2 in 17 cases.

Overall sequential dependence was evaluated using mutual information. The observed mutual information was 0.195, compared with a mean value of 0.120 under the within-whale permutation null model. None of

the 5,000 permuted datasets produced a mutual-information value as large as the observed value, resulting in a permutation p-value of approximately 0.0002.

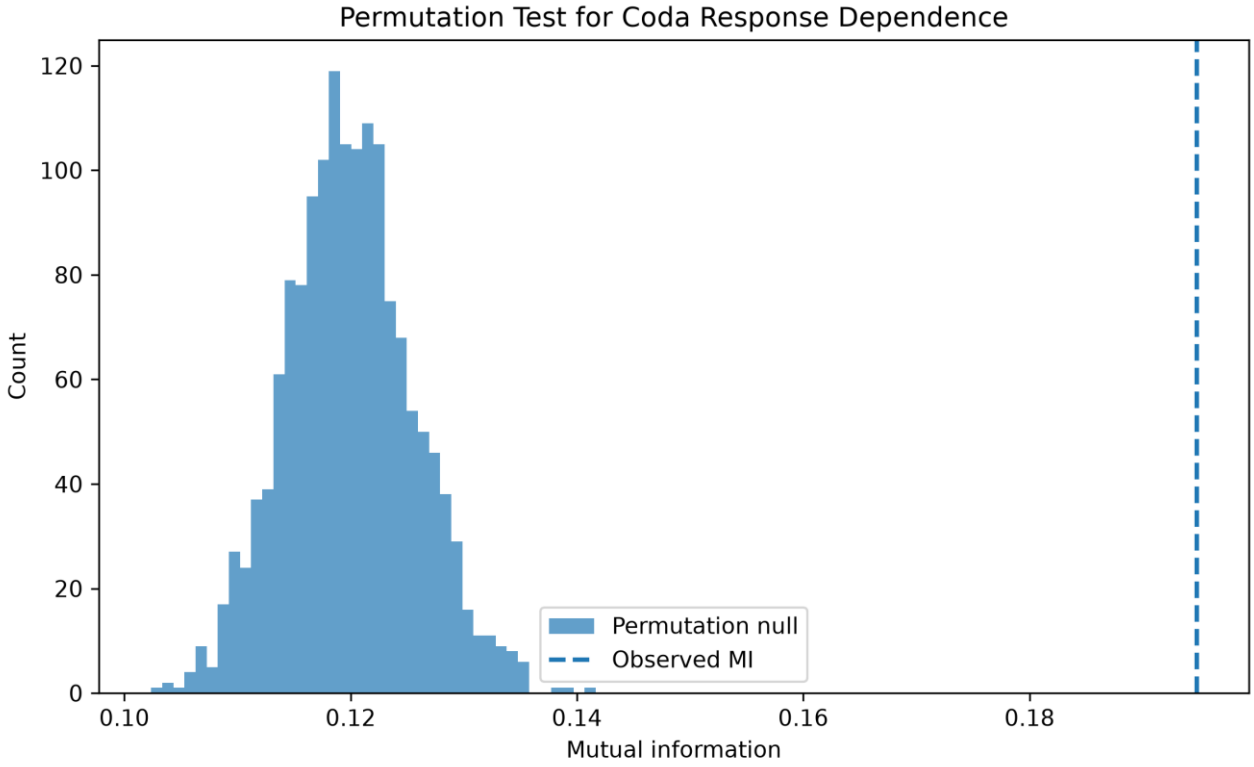


Figure 5. Permutation test for cross-speaker coda response dependence. The observed mutual information exceeded the distribution generated under the repertoire-preserving permutation null model.

Transition-specific permutation tests identified five enriched transitions after Benjamini-Hochberg FDR correction. The strongest enrichment was observed for 5R1 → 5R2, which occurred approximately 4.49 times more often than expected under the null model. The 5R2 → 5R2 and 5R2 → 5R1 transitions were enriched by approximately 2.66 and 2.52 times, respectively. Smaller but statistically significant enrichments were observed for 5R1 → 5R1 (1.12×) and 1+1+3 → 1+1+3 (1.04×). All five enriched transitions had FDR-adjusted p-values of approximately 0.0006.

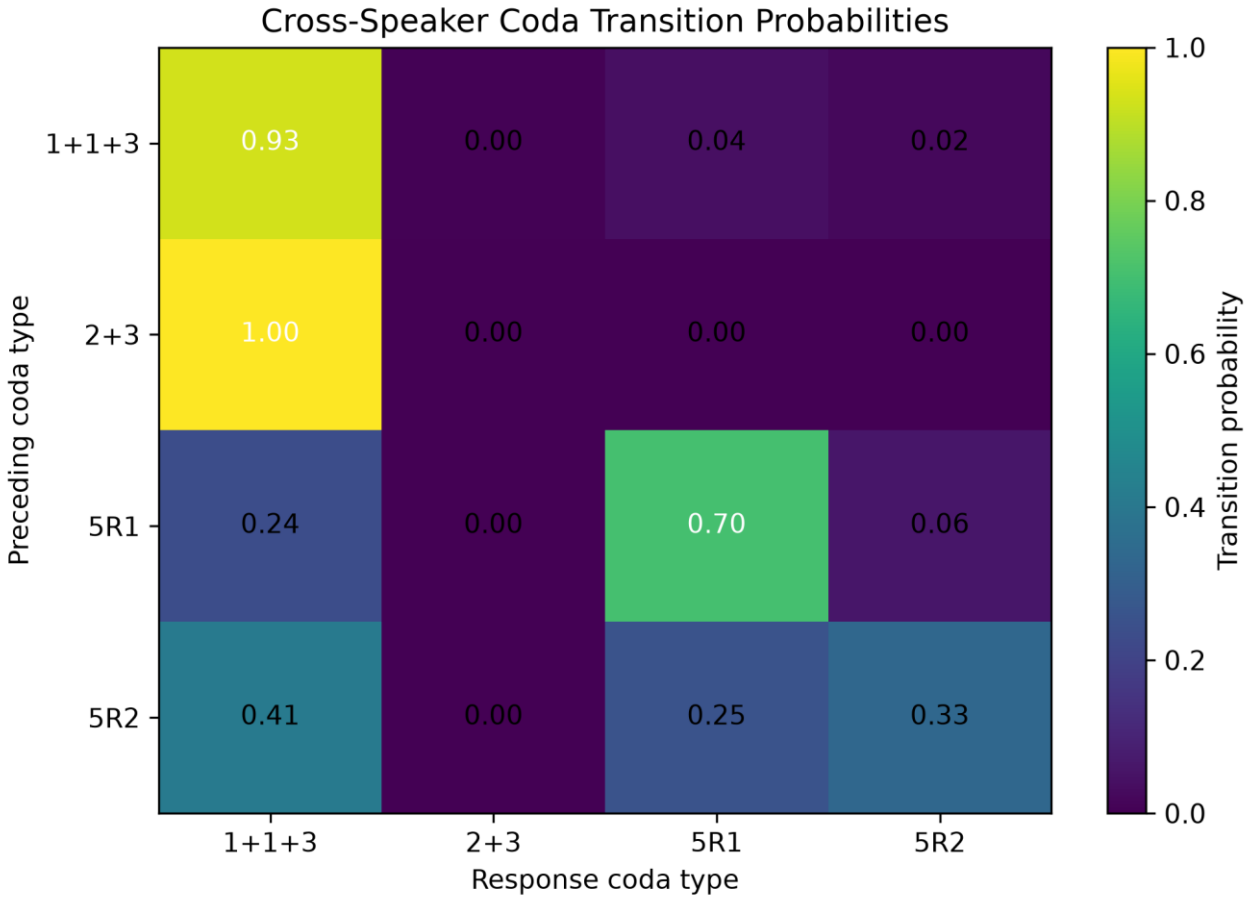


Figure 6. Cross-speaker coda transition probabilities. Rows represent preceding coda types and columns represent response coda types. Cell values show the observed transition probability within the 5-second response window.

The strongest effect sizes were therefore concentrated among transitions involving the 5R coda family. The statistically significant 1+1+3 → 1+1+3 transition had a much smaller effect size despite its large number of observations.

3.6 Robustness Checks

The main 5R transition patterns remained visible when the maximum response window was changed from 5 seconds to either 3 or 8 seconds. The number of qualifying cross-speaker pairs increased from 1,034 at the 3-second window to 1,224 at 5 seconds and 1,254 at 8 seconds.

The probability of a 5R1 coda being followed by another 5R1 remained particularly stable, accounting for approximately 69.2%, 70.4%, and 70.9% of responses following 5R1 under the 3-, 5-, and 8-second windows, respectively. The corresponding probabilities for 5R2 → 5R2 were approximately 36.1%, 33.3%, and 32.7%.

The key 5R transitions were also distributed across multiple recordings. The 119 observed 5R1 → 5R1 transitions occurred across 11 audio files, with the largest individual file contributing 22.7% of the total. The rarer 5R1 → 5R2 transition occurred across five audio files, with the largest file contributing 40.0%.

Similar results were obtained when the analysis was examined across whale-pair contexts. The 5R1 → 5R1 transition occurred in 23 pair contexts, with the largest single context contributing 11.8% of cases. The 5R2 → 5R2 transition occurred in 12 pair contexts, with the largest contributing 17.6%.

Overall, the main sequential patterns were not restricted to a single response-window definition, recording, or interacting whale pair. However, the relatively small number of 5R1 → 5R2 observations means that this transition should be interpreted more cautiously despite its strong enrichment under the permutation analysis.

4. Discussion

This study examined sperm whale communication at two related levels: the acoustic structure of individual codas and the sequential organization of codas exchanged between different whales. The results support previous evidence that sperm whale communication contains structure both within individual signals and across social exchanges, while extending the analysis toward specific cross-speaker response relationships.

The acoustic analysis showed that the selected 5-click coda types could be distinguished using relative rhythm and overall tempo. This finding is consistent with previous work demonstrating that timing features are important components of sperm whale coda structure. In the present study, acoustic classification primarily served a methodological purpose: it provided sufficiently reliable coda-type assignments for subsequent analysis of dialogue sequences rather than attempting to establish a new state-of-the-art coda classification system.

The timing results were also consistent with previous evidence of temporally coordinated coda exchanges. Speaker changes occurred after considerably shorter gaps than same-speaker continuations, and this pattern was present across all 40 audio files included in the paired comparison. These results provided a temporal foundation for examining a more specific question: whether the type of response produced by one whale was associated with the coda type that immediately preceded it.

The central result of the present analysis was that response coda type was associated with preceding coda type more strongly than expected from the responding whale's local repertoire alone. This relationship remained significant when response labels were shuffled within each AudioFile × responding Whale context, preserving local differences in coda use while disrupting sequential relationships. The result therefore suggests that the observed transition structure cannot be explained solely by individual whales preferring particular coda types.

The clearest effect-size patterns were concentrated within the 5R coda family. In particular, 5R1 → 5R2, 5R2 → 5R1, and 5R2 → 5R2 transitions occurred substantially more often than expected under the repertoire-preserving permutation model. These patterns were also observed across alternative response windows, multiple recordings, and multiple whale-pair contexts. The smaller number of some transitions,

particularly 5R1 → 5R2, requires cautious interpretation, but the results identify the 5R family as a useful candidate for further investigation of interaction-dependent coda use.

Importantly, these findings do not establish what 5R1, 5R2, or any other coda means. Statistical dependence between signals is not equivalent to semantic translation. Instead, reproducible signal-response relationships can help narrow the search for communicative function. Rather than treating thousands of codas as independent acoustic events, future studies can focus on specific exchanges and test whether they repeatedly occur in particular social or behavioral situations.

For example, future analyses could combine coda sequences with whale movement, behavioral state, social relationships, group composition, and environmental context. If a particular transition consistently occurs during the same type of interaction—for example, during changes in group association or coordinated movement—that pattern could provide stronger evidence about the communicative function of the signals involved.

In this sense, the present study should be viewed as a step toward understanding sperm whale communication rather than an attempt to decode it directly. Its main contribution is a computational framework for moving from acoustic classification to interaction-level analysis and identifying specific signal relationships that can be targeted in future behavioral research.

5. Limitations

Several limitations should be considered when interpreting these results.

First, the coda types used in the dialogue analysis were predicted by a classifier rather than provided as ground-truth annotations. A high confidence threshold and an acoustic-range filter were used to reduce uncertain assignments, but classification errors may still have influenced some transition counts.

Second, the Date + Unit grouping used during cross-validation served as a proxy for recording context and may not perfectly represent statistically independent recording sessions.

Third, the sequential analysis focused only on four selected 5-click coda types. Sperm whales produce a substantially larger repertoire of codas, and communication patterns involving other click counts or coda categories were not examined in this study.

Fourth, although the permutation procedure controlled for each responding whale's local coda repertoire within an audio file, other factors were not explicitly modeled. These may include social relationships, behavioral state, group composition, environmental conditions, and the broader conversational context surrounding each signal-response pair.

Finally, non-random sequential association does not establish semantic meaning. The results identify structured relationships between coda types, but behavioral and contextual evidence would be required before assigning communicative functions or interpretations to particular codas.

6. Conclusion

This study used acoustic analysis and sequential modeling to investigate structure in sperm whale coda communication. Selected 5-click coda types showed reproducible differences in rhythm and tempo and could be distinguished reliably across recording contexts, providing a basis for dialogue-level analysis.

Dialogue analysis revealed consistent temporal coordination between whales, while the central finding was that response coda type was associated with the preceding coda type more strongly than expected from the responding whale's local repertoire alone.

The strongest sequential effects were concentrated within the 5R coda family and remained visible across multiple response windows, recordings, and whale-pair contexts. These patterns do not reveal the semantic meaning of individual codas, but they identify specific signal relationships that may be especially informative for future behavioral and contextual research.

By narrowing a large collection of acoustic events to reproducible patterns of interaction, this approach provides a computational framework for identifying where structured communication may be occurring and for guiding future efforts to understand the communicative function of sperm whale codas.

Data and Code Availability

The two datasets analyzed in this study, `DominicaCodas.csv` and `sperm-whale-dialogues.csv`, are publicly available in the official companion repository for Sharma et al. (2024): https://github.com/pratyushasharma/sw-combinatoriality

No new acoustic recordings were collected as part of this study. The analysis code and Jupyter notebook used to produce the results and figures are available in the accompanying GitHub repository: https://github.com/msns4/whale-acoustic-ml

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